

Exact Repair-Distance Geometry of a Frozen Sixteen-Label Hénon Core

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Abstract

We give an exact finite repair atlas for the sixteen named coordinates in $Q = \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2$. For a deletion set D , the retained labels are $A = L \setminus D$, and $\rho(D)$ is the minimum number of deleted labels that must be restored before A generates the full core. Exhaustive enumeration of all $2^{16} = 65536$ deletion sets gives $\rho \in \{0, 1, 2, 3\}$ with counts 30400, 32704, 2368, 64. We record the complete bivariate polynomial $P(x, y) = \sum_D x^{|D|} y^{\rho(D)}$, where x marks deletions. Its marginals are $P(x, 1) = (1 + x)^{16}$ and $P(1, y) = 30400 + 32704y + 2368y^2 + 64y^3$. A structural derivation uses the pivot S_9 , four projective direction blocks of sizes 1, 1, 2, 5, and six dummy labels. Independent exact reconstruction, symbolic cross-check, clean replay, and hostile semantic mutations certify the receipt. The result is a finite named-coordinate statement, with no arithmetic, local, Euler-factor, root-number, Burnside-ring, or Hilbert–Polya claim.

1 Finite object and question

Let

$$Q = \mathbb{Z}/9 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/2, \quad L = \{S_1, \dots, S_{16}\}.$$

The coordinates and their closure convention are inherited byte-for-byte from the C75/C76 receipts. For a deletion set $D \subseteq L$, put $A = L \setminus D$ and define

$$\Phi(A) = \langle x_i : S_i \in A \rangle \leq Q.$$

The repair distance is

$$\rho(D) = \min\{|R| : R \subseteq D, \Phi((L \setminus D) \cup R) = Q\}. \quad (1)$$

This measures restoration cost, not the deletion count $|D|$.

The exact observable is

$$P(x, y) = \sum_{D \subseteq L} x^{|D|} y^{\rho(D)}. \quad (2)$$

Here x marks deleted labels and y marks repaired labels.

2 A structural distance formula

The preceding generation certificate identifies S_9 as a pivot and says that a retained support generates the full core precisely when it contains S_9 and meets at least two projective direction blocks. The blocks are

$$B_1 = \{S_1\}, \quad B_2 = \{S_{16}\}, \quad B_3 = \{S_7, S_{15}\}, \quad B_4 = \{S_3, S_4, S_8, S_{11}, S_{12}\}.$$

The remaining six labels $\{S_2, S_5, S_6, S_{10}, S_{13}, S_{14}\}$ are dummy labels for this criterion. Let

$$t(D) = \#\{j : (B_j \subseteq D)\}.$$

Theorem 1 (Repair formula). *For every deletion set D ,*

$$\rho(D) = \mathbf{1}_{\{S_9 \in D\}} + \max\{0, t(D) - 2\}. \quad (3)$$

In particular, $\rho(D) \leq 3$.

Proof sketch. If the pivot is deleted, one restoration is necessary and sufficient for the cyclic factor. Among the four blocks, each fully deleted block contributes a missing direction. Restoring one label in each of the excess blocks beyond the two already hit is necessary and sufficient. The pivot is disjoint from the blocks, so the costs add. The independent checker also computes the minimum directly from the complete set of 25 full-core minimal supports and verifies (3) for every mask. $\square$

3 Exact bivariate atlas

The complete coefficient table of (2) is shown below. The entries in row k are coefficients of $x^k y^0, x^k y^1, x^k y^2, x^k y^3$.

$k = D $	y^0	y^1	y^2	y^3	total
0	1	0	0	0	1
1	15	1	0	0	16
2	105	15	0	0	120
3	455	105	0	0	560
4	1364	456	0	0	1820
5	2992	1375	1	0	4368
6	4950	3047	11	0	8008
7	6269	5116	55	0	11440
8	6095	6609	166	0	12870
9	4504	6595	341	0	11440
10	2461	5040	506	1	8008
11	940	2871	551	6	4368
12	224	1151	430	15	1820
13	25	289	226	20	560
14	0	34	71	15	120
15	0	0	10	6	16
16	0	0	0	1	1

The row totals are $\binom{16}{k}$, giving

$$P(x, 1) = (1 + x)^{16}.$$

The column totals give the repair distribution

$$P(1, y) = 30400 + 32704y + 2368y^2 + 64y^3.$$

The table also has a compact derivation. Define

$$H(x, z) = \prod_{s \in \{1, 1, 2, 5\}} \left(\sum_{d=0}^{s-1} \binom{s}{d} x^d + z x^s \right). \quad (4)$$

The variable z records the number of fully deleted blocks. Replacing each z^r by $y^{\max(0, r-2)}$, and then adding the dummy and pivot factors, gives

$$P(x, y) = (1 + x)^6 (1 + xy) \text{ Transform}_{z \rightarrow y}(H(x, z)). \quad (5)$$

Expanding (5) agrees coefficient-by-coefficient with direct enumeration.

4 Certification and scope

The producer binds the C73 generation criterion, C75 coordinate source, C76 closure atlas, and C77 subgroup receipt by SHA-256 before computing. An independent checker reconstructs the point-set closure transitions and the 25 full-core minimal supports, derives the four blocks from the coordinates, and checks all 65536 masks. A symbolic cross-check verifies (5); a fresh-process replay preserves the canonical evidence bytes; and a hostile audit rejects 19 semantic mutations.

The canonical evidence hash is 728d6462b337e3b22fe267ae9388da476a0f6409cc64a17ca659f53f1a8126ae. The effective C76 label action and the preceding lifted ambient action remain source-bound distinctions, but no group-orbit assertion is needed for (2). Scope is explicitly limited to this finite named presentation: there is no arithmetic/local, Euler-factor, root-number, automorphy, full Burnside-ring/table-of-marks, or Hilbert–Polya claim.

5 Conclusion

The repair geometry of the frozen support is shallow: every deletion pattern can be repaired with at most three restored labels, and the exact joint atlas is determined by one pivot and four direction blocks. The result supplies a reproducible finite certificate for subsequent combinatorial work while preserving the source and interpretation boundaries of the preceding papers.